



**BAULKHAM HILLS HIGH SCHOOL**

**Assessment Task 2, 2016**  
**YEAR 11**

# **Mathematics Extension 1**

## **General Instructions**

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 1-4
- Marks may be deducted for careless or badly arranged work

**Total marks – 41**

**Exam consists of 3 pages.**

This paper consists of 4 Questions.

**QUESTION 1 (9 marks)**

- a) Given the points  $A(-5,6)$  and  $B(-2,3)$ , find the point  $P$  which divides the interval  $AB$  externally in the ratio  $3:2$ . 2
- b) Find the exact value of  $\sin 255^\circ$ . 3
- c) Solve  $\cos 2\theta + \sin 2\theta = 2\cos^2\theta$  for  $0^\circ \leq \theta \leq 360^\circ$ . 2
- d) If  $\sin x = \frac{1}{3}$  and  $\cos y = \frac{2}{3}$ , and  $x$  and  $y$  are acute angles, evaluate  $\cos(x - y)$  2

**QUESTION 2 (9 marks)**

- a) The point  $P(1, y)$  divides the interval  $AB$  internally in the ratio  $m:1$ , where  $A = (-5,2)$  and  $B = (3, -1)$ .
- (i) Find the value of  $m$ . 2
- (ii) What is the  $y$ -coordinate of  $P$ ? 1
- b) Given  $\sin 18^\circ = \frac{1}{4}(\sqrt{5} - 1)$ , find the exact value of  $\cos 36^\circ$  in simplest form. 2
- c) (i) Express  $\sqrt{3} \cos x - \sin x$  in the form  $R \cos(x + \alpha)$  2
- where  $R > 0$  and  $0^\circ < \alpha < 90^\circ$ .
- (ii) Hence or otherwise, solve  $\sqrt{3} \cos x - \sin x = \sqrt{3}$  for  $0^\circ \leq x \leq 360^\circ$ . 2

**QUESTION 3 (11 marks)**

a) The line  $y = mx$  makes an angle of  $30^\circ$  with the line  $6x - 2y + 7 = 0$ . Find the possible values of  $m$  in exact form.

**3**

b) If  $\tan \frac{\theta}{2} = \frac{a}{b}$ , express  $b \cos \theta + a \sin \theta$  in its simplest form.

**3**

c) (i) Prove

$$\frac{\sin 2A}{1 - \cos 2A} = \cot A$$

**2**

(ii) Hence find integers  $a$  and  $b$  such that  $\cot 67.5^\circ = a + \sqrt{b}$

**3****QUESTION 4 (12 marks)**

a)  $A, B$  and  $C$  are three acute angles whose sum is  $45^\circ$ .

**3**

If  $\tan A = 0.5$  and  $\tan B = 0.25$ , find the exact value of  $\tan C$ .

b) Prove that

$$\frac{\sin x + \sin(x+2y)}{2 \cos y} = \sin(x+y)$$

**3**

c) (i) Prove that  $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$

**2**

(ii) Hence or otherwise solve  $\sin 2\theta = \cos 3\theta$  for  $-180^\circ \leq \theta \leq 180^\circ$ .

**4****End of Exam**

Question 1.

TOTAL/41

a)  $A(-5, 6)$   $B(-2, 3)$

$3: -2$

$$x = \frac{3(-2) - 2(-5)}{3 - 2}$$

$= 4$

$$y = \frac{3(3) - 2(6)}{3 - 2}$$

$= -3$

$\therefore P = (4, -3)$

(1) x-value

(1) y-value

[For internal, max 1 mark]

b)  $\sin 255^\circ = -\sin 75^\circ$   $\leftarrow (1)$

$= -\sin(45^\circ + 30^\circ)$

$= -(\sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ)$   $\leftarrow (1)$

$= -\left(\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\right)$

$= -\left(\frac{\sqrt{3} + 1}{2\sqrt{2}}\right)$  or  $\frac{-\sqrt{3} - 1}{2\sqrt{2}}$   $\leftarrow (1)$

c)  $2\cos^2 \theta - 1 + \sin 2\theta = 2\cos^2 \theta$   $\left[ \begin{array}{l} 0 \leq \theta \leq 360^\circ \\ 0 \leq 2\theta \leq 720^\circ \end{array} \right]$

$\sin 2\theta = 1$   $\leftarrow (1)$

$2\theta = 90^\circ, 450^\circ$

$\theta = 45^\circ, 225^\circ$   $\leftarrow (1)$

d)  $\cos(x-y) = \cos x \cos y + \sin x \sin y$

$= \frac{\sqrt{8}}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{\sqrt{5}}{3}$   $\leftarrow (1)$

$= \frac{2\sqrt{8} + \sqrt{5}}{9}$

$= \frac{4\sqrt{2} + \sqrt{5}}{9}$

$\leftarrow (1)$

## Question 2

a) i)  $A(-5, 2)$   $B(3, -1)$   $P(1, y)$   
 ~~$P(1, y)$~~

$$1 = \frac{m(3) + 1(-5)}{m+1} \quad \leftarrow (1)$$

$$m+1 = 3m-5$$

$$6 = 2m$$

$$m = 3$$

$\leftarrow (1)$

ii)  $y = \frac{3(-1) + 1(2)}{3+1} = -\frac{1}{4} \quad \leftarrow (1)$

b)  $\cos 36^\circ = 1 - 2 \sin^2 18^\circ \quad \leftarrow (1)$

$$= 1 - 2 \cdot \frac{1}{16} (\sqrt{5}-1)^2$$

$$= 1 - \frac{1}{8} (6 - 2\sqrt{5})$$

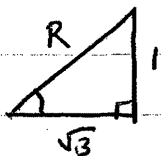
$$= \frac{1}{4} + \frac{\sqrt{5}}{4} \quad \text{or} \quad \frac{1+\sqrt{5}}{4} \quad \leftarrow (1)$$

c) (i)  $\sqrt{3} \cos x - \sin x = R \cos(x+\alpha)$

$$= R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$R \cos \alpha = \frac{\sqrt{3}}{R}$$

$$R \sin \alpha = \frac{1}{R}$$



$$\therefore R = 2 \quad \leftarrow (1)$$

$$\alpha = 30^\circ \quad \leftarrow (1)$$

$$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x+30^\circ)$$

(ii)  $\cancel{2} \cos(x+30^\circ) = \frac{\sqrt{3}}{2} \quad \left[ \begin{array}{l} 0 \leq x \leq 360^\circ \\ 30^\circ \leq x \leq 390^\circ \end{array} \right]$

$$x+30^\circ = 30^\circ, 330^\circ, 390^\circ$$

$$x = 0^\circ, 300^\circ, 360^\circ \quad \leftarrow (2)$$

$\left( \begin{array}{l} 2 \text{ correct} = 1 \text{ mark} \\ 1 \text{ correct} = 0 \text{ mark} \end{array} \right)$

### Question 3

$$a) \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 30^\circ = \left| \frac{m - 3}{1 + 3m} \right|$$

$$\frac{1}{\sqrt{3}} = \frac{|m - 3|}{|1 + 3m|} \quad \leftarrow (1)$$

$$\begin{aligned} \sqrt{3}(m - 3) &= 1 + 3m & \text{or} & \quad \sqrt{3}(m - 3) = -1 - 3m \\ (\sqrt{3} - 3)m &= 1 + 3\sqrt{3} & & \quad (\sqrt{3} + 3)m = -1 + 3\sqrt{3} \end{aligned}$$

$$m = \frac{1 + 3\sqrt{3}}{\sqrt{3} - 3} \quad \leftarrow (1)$$

$$m = \frac{-1 + 3\sqrt{3}}{\sqrt{3} + 3} \quad \leftarrow (1)$$

$$b) \text{ Using } t = \tan \frac{\theta}{2},$$

$$b \cos \theta + a \sin \theta = b \left( \frac{1 - t^2}{1 + t^2} \right) + a \left( \frac{2t}{1 + t^2} \right) \quad \leftarrow (1)$$

$$= \frac{b(1 - t^2) + 2at}{1 + t^2}$$

$$= \frac{b - bt^2 + 2at}{1 + t^2}$$

$$= \frac{b - b \cdot \frac{a^2}{b^2} + 2a \cdot \frac{a}{b}}{1 + \frac{a^2}{b^2}} \quad \leftarrow (1)$$

$$= \frac{b - \frac{a^2}{b} + \frac{2a^2}{b}}{1 + \frac{a^2}{b^2}}$$

$$= \frac{b + \frac{a^2}{b}}{1 + \frac{a^2}{b^2}} \times \frac{b^2}{b^2}$$

$$= \frac{b^3 + a^2 b}{b^2 + a^2}$$

$$= \frac{b(b^2 + a^2)}{b^2 + a^2}$$

$$= b.$$

← (1)

$$c) (i) \text{ LHS} = \frac{\sin 2A}{1 - \cos 2A}$$

$$= \frac{2 \sin A \cos A}{1 - (1 - 2 \sin^2 A)}$$

$$= \frac{\cancel{2} \sin A \cos A}{\cancel{2} \sin^2 A}$$

$$= \frac{\cos A}{\sin A}$$

$$= \cot A = \text{RHS.}$$

$$(ii) \cot 67.5^\circ = \frac{\sin 135^\circ}{1 - \cos 135^\circ}$$

$$= \frac{\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \times \frac{\sqrt{2}}{\sqrt{2}} \quad (1)$$

$$= \frac{1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$$

$$= \sqrt{2} - 1$$

$$\therefore a = -1$$

← (1)

$$b = 2$$

← (1)

Question 4.

a)  $\tan((A+B)+C) = \tan 45^\circ$

$$\frac{\tan(A+B) + \tan C}{1 - \tan(A+B)\tan C} = 1 \quad \leftarrow (1)$$

$$\text{Now } \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{0.5 + 0.25}{1 - 0.5 \times 0.25}$$

$$= \frac{0.75}{0.875}$$

$$= \frac{6}{7}$$

$$\leftarrow (1)$$

$$\frac{\frac{6}{7} + \tan C}{1 - \frac{6}{7} \cdot \tan C} = 1$$

$$\frac{6}{7} + \tan C = 1 - \frac{6}{7} \tan C$$

$$\frac{13}{7} \tan C = \frac{1}{7}$$

$$\tan C = \frac{1}{13}$$

$$\leftarrow (1)$$

b) LHS =  $\frac{\sin x + \sin(x+2y)}{2 \cos y}$

$$= \frac{\sin x + \sin x \cos 2y + \cos x \sin 2y}{2 \cos y}$$

$$= \frac{\sin x (1 + \cos 2y) + \cos x \cdot 2 \sin y \cos y}{2 \cos y} \quad \leftarrow (1)$$

$$= \frac{\sin x \cdot 2 \cos^2 y + 2 \cos x \sin y \cos y}{2 \cos y}$$

$$\begin{cases} \cos 2y = 2 \cos^2 y - 1 \\ \therefore 1 + \cos 2y = 2 \cos^2 y \end{cases}$$

$$= \frac{2 \cos y (\sin x \cos y + \cos x \sin y)}{2 \cos y}$$

$$= \sin x \cos y + \cos x \sin y$$

← (1)

$$= \sin(x+y)$$

$$= \text{RHS}$$

$$c) i) \cos 3\theta = \cos(2\theta + \theta)$$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= (2\cos^2\theta - 1)\cos\theta - 2\sin\theta \cos\theta \cdot \sin\theta \quad \leftarrow (1)$$

$$= 2\cos^3\theta - \cos\theta - 2(1 - \cos^2\theta) \cdot \cos\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta \quad \leftarrow (1)$$

$$= 4\cos^3\theta - 3\cos\theta$$

$$ii) \sin 2\theta = \cos 3\theta$$

$$2\sin\theta \cos\theta = 4\cos^3\theta - 3\cos\theta$$

$$4\cos^3\theta - 3\cos\theta - 2\sin\theta \cos\theta = 0$$

$$\cos\theta \left( \underbrace{4\cos^2\theta}_{1-\sin^2\theta} - 2\sin\theta - 3 \right) = 0$$

$$\cos\theta (4 - 4\sin^2\theta - 2\sin\theta - 3) = 0$$

} ← (1)

$$\cos\theta (-4\sin^2\theta - 2\sin\theta + 1) = 0$$

$$\therefore \cos\theta = 0$$

or

$$-4\sin^2\theta - 2\sin\theta + 1 = 0$$

$$4\sin^2\theta + 2\sin\theta - 1 = 0$$

$$\sin\theta = \frac{-2 \pm \sqrt{4+16}}{8}$$

$$= \frac{-2 \pm \sqrt{20}}{8}$$

$$\therefore \theta = 0.3090, -0.8090$$

$$\therefore \theta = \underline{90^\circ}, \underline{-90^\circ}, \underline{18^\circ}, \underline{162^\circ}, \underline{-54^\circ}, \underline{-126^\circ}$$